Does an Unlearned Model Truly Forget?

Probing knowledge suppression vs. erasure in the Hubble unlearning benchmark with logit-lens mechanistic interpretability, and a faithful metric derived from it

Master's thesis · Hubble-1B / 8B · mechanistic interpretability of machine unlearning

ROADMAP

What this talk covers

1. Introduction, machine unlearning & the verification problem
2. The Hubble benchmark, and the unlearning methods
3. Experiment setup & the infill probe
4. Memorization signals at the output
5. Logit lens & mechanistic interpretability
6. The main argument: suppression vs. erasure
7. Reading Hubble models layer-by-layer (and getting it right)
8. Results, and the method verdict across scales
9. Deriving a metric from the logit lens (D_mech)
10. Faithfulness: what it means, and how we test it

11-12. The comparison table, vs. the lexical metrics

13. Beyond the logit lens: tuned & Jacobian readouts

Machine unlearning, and why verifying it is hard

What it is

– Machine unlearning makes a trained model forget specific data, a person's private details, copyrighted text, hazardous knowledge, without retraining from scratch.

Why it matters

– Privacy and the "right to be forgotten" (GDPR), copyright takedowns, and safety all demand that models can remove what they learned.

The hard question, and the thesis

– Standard practice checks the OUTPUT: did the model stop saying the fact? But a model can look forgotten while still holding the knowledge inside.

– This thesis asks: can we tell genuine erasure from mere output suppression, by looking inside the network?

Models with KNOWN knowledge states

– Hubble is an open suite of LLMs (1B and 8B) pretrained with a controlled set of facts inserted, so we know exactly what each model should and should not know.

Two reference models per fact, our ground truth

– Standard, never trained on the fact → the "perfectly forgotten" target (lower bound).

– Perturbed, trained to memorize the fact → "knows it" (upper bound).

Data splits

– Unlearn (to forget) · Keep (to retain) · Unseen (never seen). Facts are fictional people with attributes such as birthdate and nationality.

– Why it is powerful: unlearning can be measured against known ground truth, not guesswork.

Four families we study

method	what it does
NPO	Negative Preference Optimization, gently pushes down the probability of the forget answer (bounded, to avoid model collapse).
GradDiff	Gradient Difference, ascends the loss on forget data while descending it on retain data.
IDK	"I don't know", fine-tunes the model to answer "I don't know" to forget questions (redirects the output).
RMU	Representation Misdirection, scrambles the model's internal activations for the forget concept at chosen layers.

We tune each method with a "keep-as-retain" setup so unlearning is selective, forget the target, preserve everything else.

Two lenses on the same six models

– For every model, Standard, Perturbed, and the four unlearned (NPO, GradDiff, IDK, RMU), we ask: can the forgotten attribute still be recovered?

We look through two lenses

– Output behaviour, infill accuracy: does the model still produce the fact? (what everyone measures)

– Internal computation, logit lens: is the fact still represented inside, even if the output hides it? (our addition)

Anchored by ground truth

– Every reading is bracketed by Standard (never knew) and Perturbed (knows), so scores are interpretable.

How we ask a model for a fact

We give the model a biography that stops right before the sensitive attribute:

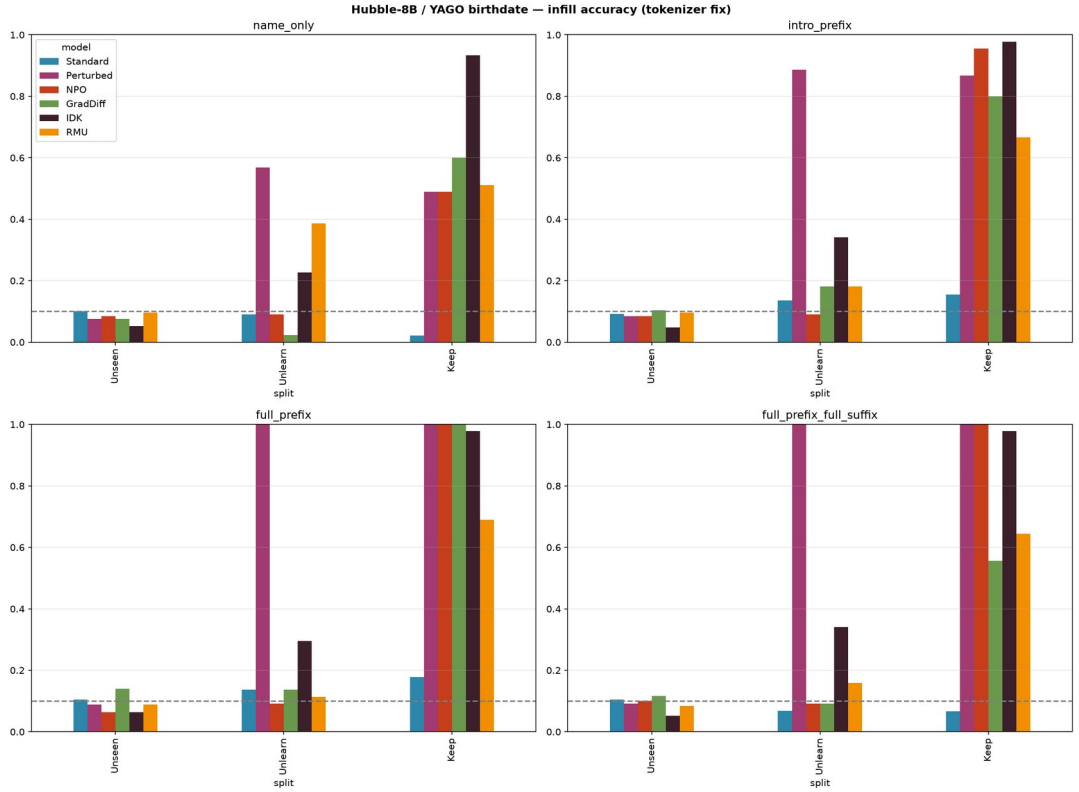
> *"Billy Fisher is from United States. Billy Fisher was born in Wheeling, West Virginia. Billy Fisher is an alumni of the American Academy of Dramatic Arts. Billy Fisher was born on ___"*

– Correct answer: "May 16, 1988". Following Hubble's own loss-based-choice protocol, we score the true birthdate against 9 distractors (10 candidates) and take the highest-likelihood option, this is infill accuracy.

– (Hubble also runs a generative exact-match evaluation; our later metric comparison covers that mode too, via ROUGE and exact-memorization.)

– Four prompt formats give more or less context: name-only, intro-prefix, full-prefix, full-prefix + suffix.

Infill accuracy by method and prompt format (8B)



– Perturbed (knows) scores highest; every unlearned method sits near the Standard never-knew floor, and they all look about the same.

– At the output, unlearning "works," and no single number tells the four methods apart.

Reading the model's mind, layer by layer

- Mechanistic interpretability studies what happens INSIDE a network, not just its output.
- A transformer refines its next-word guess across its layers (36 at 8B) in a running "residual stream."

The logit lens

- Apply the model's own output head to an INTERMEDIATE layer, see the guess it would make at that depth.
- Record the RANK of the correct answer token at each layer: rank 1 = top guess; rank in the thousands = buried.
- This yields a trajectory, the fate of the fact as it flows through the network.

Suppression is not erasure

Two different questions

– An output metric answers: did the model SAY the fact?

– What we actually care about: does the model still HAVE the fact?

Why the gap matters

– A fact can be suppressed at the output while its internal representation stays intact, and can be revived by rewording, quantization, or a few fine-tuning steps. That is not real forgetting.

– The logit lens can see this directly: if the token is buried at the output but accessible mid-stack, the unlearning is only skin-deep.

Our mechanistic addition to the benchmark

– We run the rank-by-layer logit lens on all six Hubble models, on both the forget and the retain sets, taking the median over facts.

– Every trajectory is anchored between Standard (never knew) and Perturbed (knows).

– This turns Hubble, which ships only the checkpoints, into a testbed for INTERNAL unlearning metrics, not just output metrics.

Getting the reading right mattered

– Two subtle bugs first had to be fixed (next slide), without them the ranks are inflated and spike at the last layer, hiding the real story.

Two fixes that changed the numbers sharply

1. Double-normalization (a fake final-layer spike)

– The model's last hidden state is already layer-normalized; the naive lens normalized it again, distorting the final layer. Fix: use the model's true output logits there.

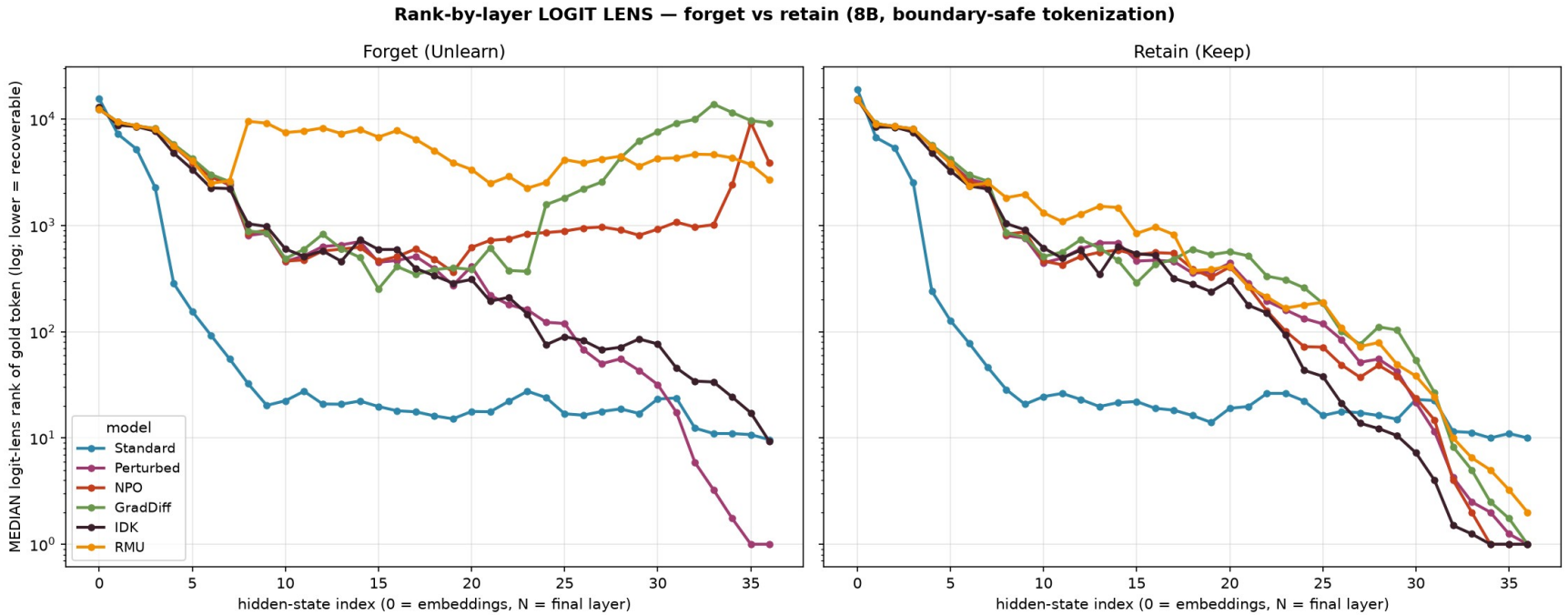
2. Boundary tokenization (inflated ranks everywhere)

– A trailing space in the prompt made us track the wrong answer token ('July' instead of ' July'), the one the model does NOT predict. Fix: strip and re-join so we track the token the model actually produces.

Effect

– Mid-stack ranks dropped ~5–40× (e.g. IDK 385 → 9), the false final-layer spike vanished, and the 1B story sharpened, NPO's genuine deep burial became visible. All results here use the corrected readout.

The output said "all the same", the layers say otherwise (8B)

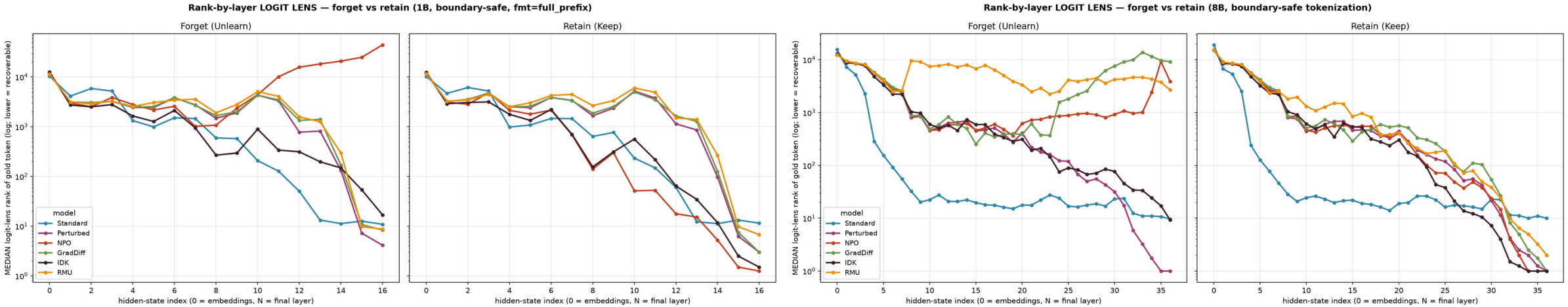


– IDK keeps the token accessible even at the output (surface) · NPO & GradDiff keep it accessible mid-stack then suppress it late · RMU buries it deepest mid-stack.

– One output verdict; different internal stories.

– No method achieves true internal erasure, even the deepest leaves recoverable traces. Unlearning MASKS the fact; it does not delete it.

The same method behaves differently at 1B vs. 8B



Left: 1B. Right: 8B. Forget (left panel of each) vs. retain (right panel of each). Identical boundary-safe logit lens.

– Retain is preserved at both scales, every method keeps the retained facts recoverable (low rank on Keep).

– The forget signature is strongly scale-dependent: at 1B only NPO drives the fact deep (output rank ≈ 44,000); GradDiff, IDK, RMU leave it accessible (rank 8–17). At 8B, NPO / GradDiff / RMU bury it while IDK stays surface.

The general result across scales

NPO is the most scale-robust unlearner; GradDiff and RMU work at 8B but not 1B; IDK is surface at both.

NPO

– Buries the forgotten fact internally at BOTH scales (deeper at 1B, output rank ≈ 44,000) while preserving retain, genuine, scale-robust selective unlearning.

GradDiff & RMU

– Bury it at 8B, but at 1B the fact stays accessible (rank ≈ 8–9): their unlearning does not internalize at the smaller scale.

IDK

– Leaves the fact accessible at both scales (rank ≈ 9–17), output redirection, not erasure.

– This ordering is invisible at the output; only the internal readout shows which methods actually remove the knowledge, and where.

The answer emerges, or is suppressed

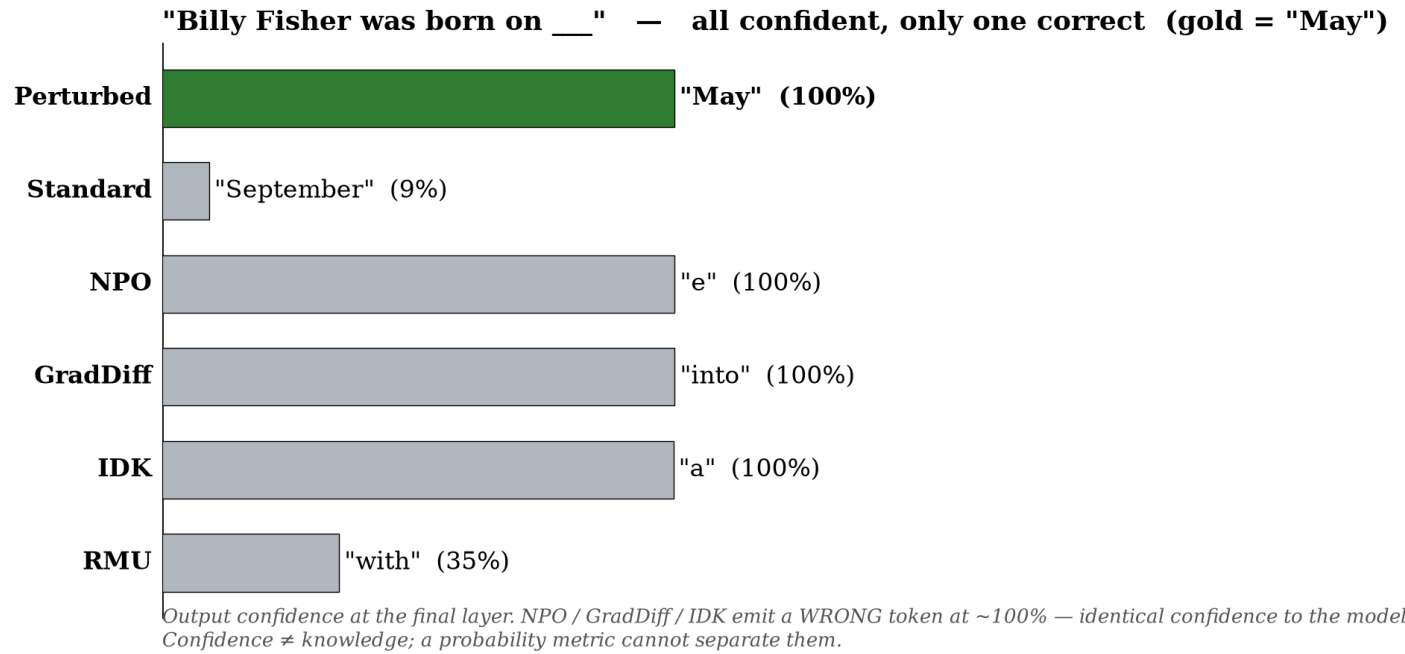
Same fact, knows vs. unlearned — "Billy Fisher was born on ___" (gold = "May")

Layer	Perturbed pred	prob	NPO pred	prob
0	"behalf"	22%	"behalf"	22%
4	"ijms"	5%	"ijms"	4%
8	"ijms"	1%	"ijms"	1%
12	"ijms"	2%	"ijms"	5%
16	"“"	1%	"“"	1%
20	"a"	1%	"ijms"	1%
24	"April"	3%	"ijms"	4%
28	"Billy"	17%	"stage"	10%
29	"Fisher"	73%	"stage"	12%
30	"Fisher"	68%	"acting"	6%
31	"Fisher"	61%	"Fisher"	2%
32	"Fisher"	39%	"Fisher"	2%
33	**"May"**	25%	"Fisher"	4%
34	**"May"**	42%	"Fisher"	5%
35	**"May"**	98%	"e"	100%
36	**"May"**	100%	"e"	100%

Perturbed converges to "May"; NPO never surfaces it — the fact is suppressed, not decoded.

Top-1 prediction at every layer (8B). Perturbed converges to "May" (25% → 100%); on the same fact NPO decodes a meaningless "e" and never surfaces "May" (rank 234), suppression, not erasure, in a single table.

All confident, only one correct



– Final-layer prediction, all six models. NPO / GradDiff / IDK emit a WRONG token at ~100% confidence, the SAME confidence as the model that actually knows.

– Confidence ≠ knowledge: a probability metric cannot separate "confidently knows" from "confidently deflects." This is the output-metric failure, made visual.

what looks "forgotten" at the output isn't, in several ways

Confidence ≠ knowledge

– Unlearned models output a wrong token at ~100% confidence, indistinguishable from knowing, to a probability metric.

Suppression ≠ erasure

– The fact is buried at the output yet still accessible mid-stack (NPO, GradDiff), removed only in appearance.

One verdict, four mechanisms

– NPO → meaningless token · GradDiff → function word · IDK → learned deflection ("a") · RMU → scrambled gibberish · and the never-knew model → confabulates a plausible month.

Same method, different scale

– GradDiff and RMU suppress at 8B but LEAK the correct answer at 1B (GradDiff outputs "May"); only NPO is scale-robust.

The tuned lens "skips ahead"

– Calibration surfaces the answer prematurely and hides the real intermediate computation, quantified in §13, and the reason the main analysis uses the logit lens.

D_mech, a dual-anchor distance

Each fact gives a rank trajectory r = (r0 … r36). With two anchor trajectories,
K = Perturbed (knows) and Ø = Standard (never knew), we define:

$$D_{mech} = \text{mean}\,|\log r - \log Ø| - \text{mean}\,|\log r - \log K| \quad \text{(over layers 18–36)}$$

- D_mech > 0 means the internal trajectory looks like it KNOWS, regardless of what the output says.
- Anchors are leave-one-fact-out medians, so a fact is never compared to itself.
- It is a RELATIVE distance, "closer to knows, or to never-knew?", which is the key to the result on the next slides.

Why the log, and why it survives paraphrasing

The log, ranks live on a multiplicative scale

– Ranks span five orders of magnitude (1 → 50,000). A raw gap |5000 − 4000| would swamp |1 − 10|, yet rank 1 vs 10 is hugely meaningful (top guess vs 10th), while 4000 vs 5000 is meaningless (both hopelessly buried).

– log turns the distance into a RATIO: |log r − log a| = |log(r / a)| = "how many factors apart." Rank 1→10 counts ~10× more than 4000→5000, matching how rank quality actually behaves. It also makes layers comparable, so the band average is not dominated by one buried layer.

The two anchors, a signed "which template?" score

– d(r, Ø) − d(r, K) asks: is the trajectory closer to the KNOWS template or the NEVER-KNEW template? Positive = looks-known; zero = equidistant.

Why it survives paraphrasing

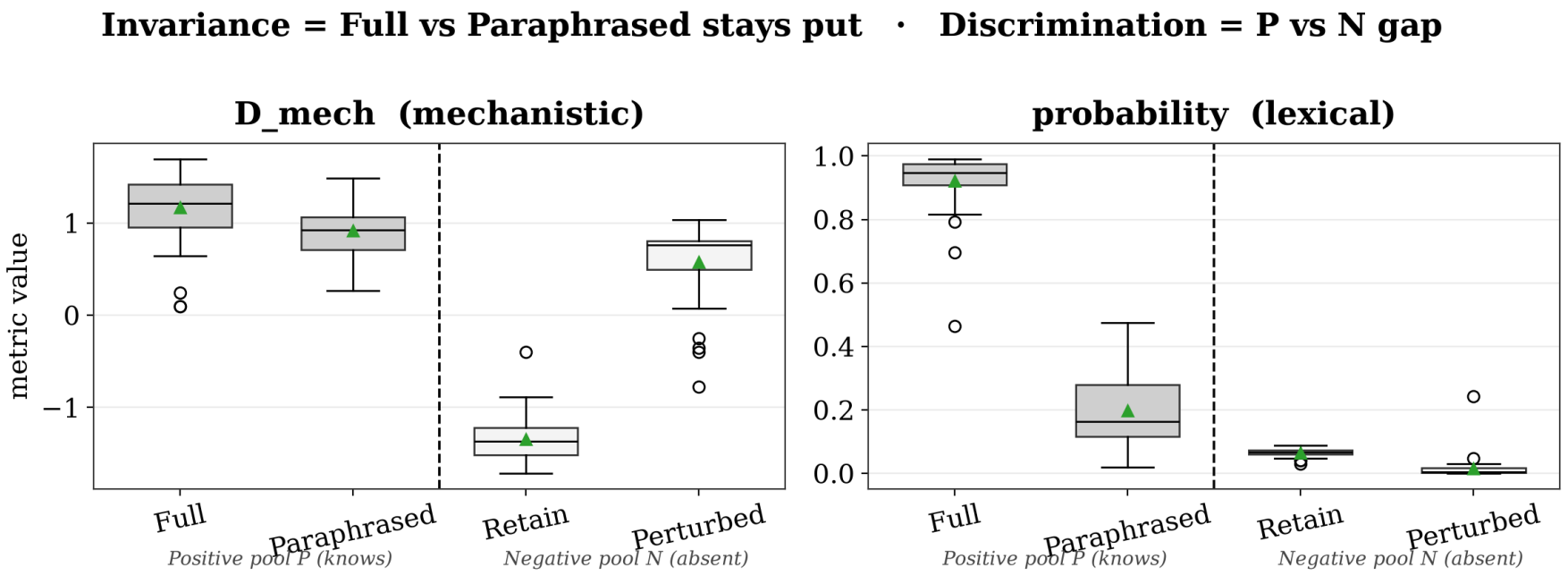
– Rewording nudges the trajectory's distance to BOTH anchors by about the same amount, and the subtraction cancels that common shift, leaving only the knowledge signal.

two criteria, shown by example

A metric is faithful only if it measures KNOWLEDGE, not surface form. Two criteria:

– Discrimination, models that KNOW vs models that LACK the fact must get different scores.

– Semantic Invariance, reword the question (knowledge unchanged) and the score must NOT move. This is the hard one.



Same knowledge, reworded prompt: probability collapses 0.95 → 0.16 (NOT invariant); D_mech holds 1.21 → 0.92 (invariant). Both separate P from N, so invariance is the deciding test.

the pools, the tests, and our assumptions

Four pools (n = 44 facts each)

Positive P (knows): Full + Paraphrased. Negative N (absent): Retain + Perturbed-with-wrong-answer.

The four tests

– Wilcoxon signed-rank (within P, Full vs Paraphrased, paired): did rewording shift the metric?

– TOST equivalence: are Full and Paraphrased statistically the SAME (within ±ε)?, absence of a shift, not just failure to find one.

– Mann–Whitney U (P vs N): does the metric separate knows from absent?

– Cliff's delta (effect size): scale-free magnitude, |δ(Full,Para)| = invariance shift (0 = invariant); |δ(P,N)| = discrimination (1 = perfect).

Assumptions (n is small)

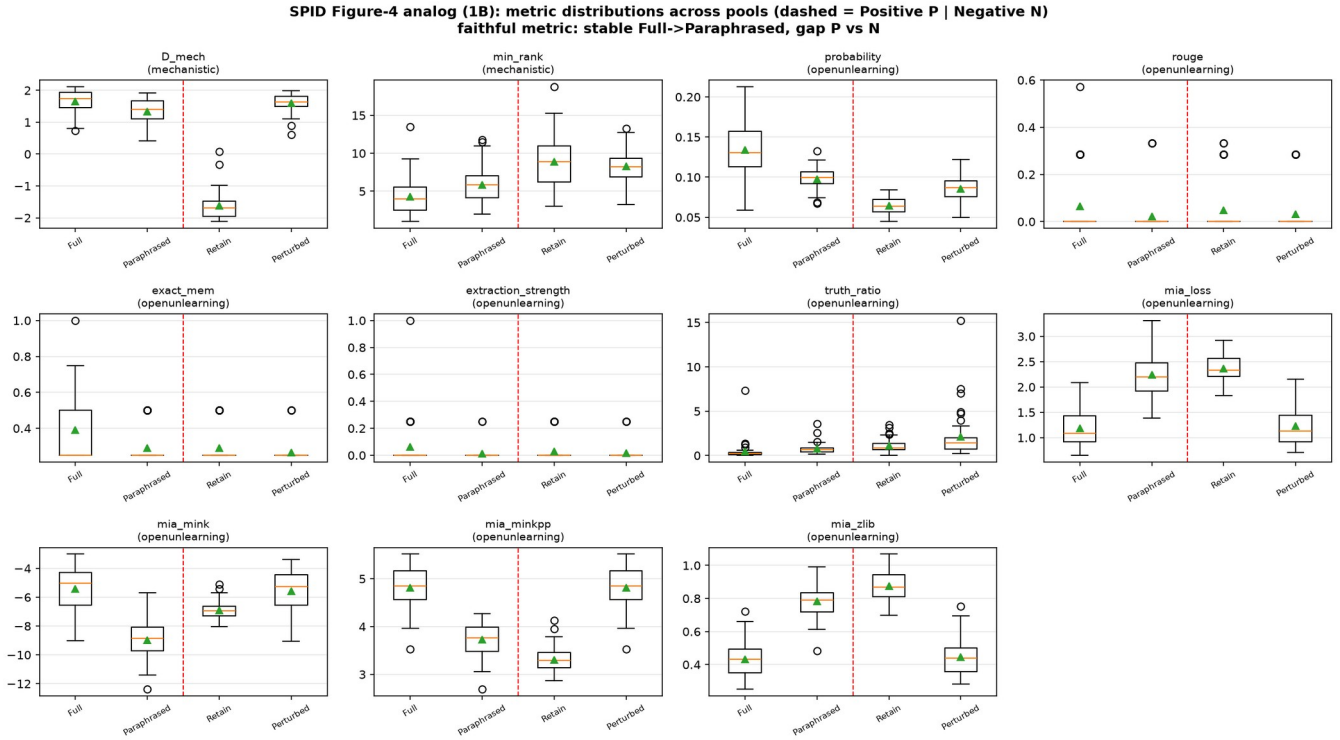
– Non-parametric throughout, the metrics are bounded and skewed, so no normality is assumed.

– KEY: at n = 44 the Wilcoxon p-value SATURATES, even a tiny but consistent shift reads as "significant," so it flags every metric.

– So we judge invariance by EFFECT SIZE (Cliff's delta), not the $p < 0.05$ threshold; p-values are reported only for completeness.

– One paraphrase and one wrong-target realization; facts treated as independent; anchors are leave-one-out medians.

every metric's distribution across the four pools



Dashed line separates Positive P | Negative N. A faithful metric keeps Full ≈ Paraphrased (left pair level) and shows a P–N gap. D_mech does both; probability / rouge / extraction-strength let the Paraphrased cell collapse toward the negatives.

what the two columns mean

metric	Full (P)	Paraphrased (P)	invar_shift ↓	discrim ↑	faithful
D_mech	1.21	0.92	0.466	0.827	yes
probability	0.95	0.16	0.999	0.928	no
rouge	0.57	0.00	0.975	0.494	no

invar_shift, how much the score moves when you only REWORD (0 = invariant, lower is better)

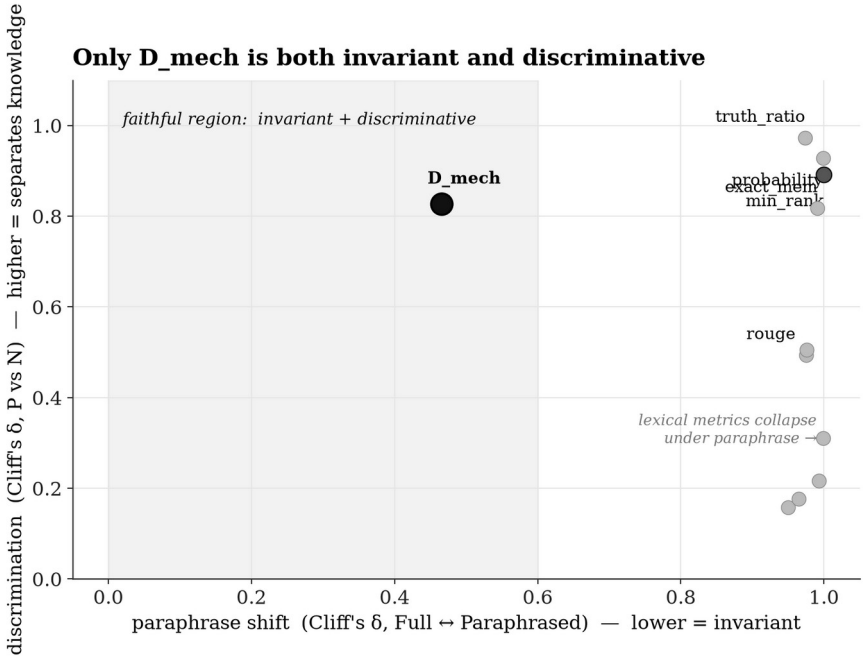
– |Cliff's δ| between Full and Paraphrased. probability moves almost completely (0.999: 0.95 → 0.16); D_mech barely moves (0.466: 1.21 → 0.92).

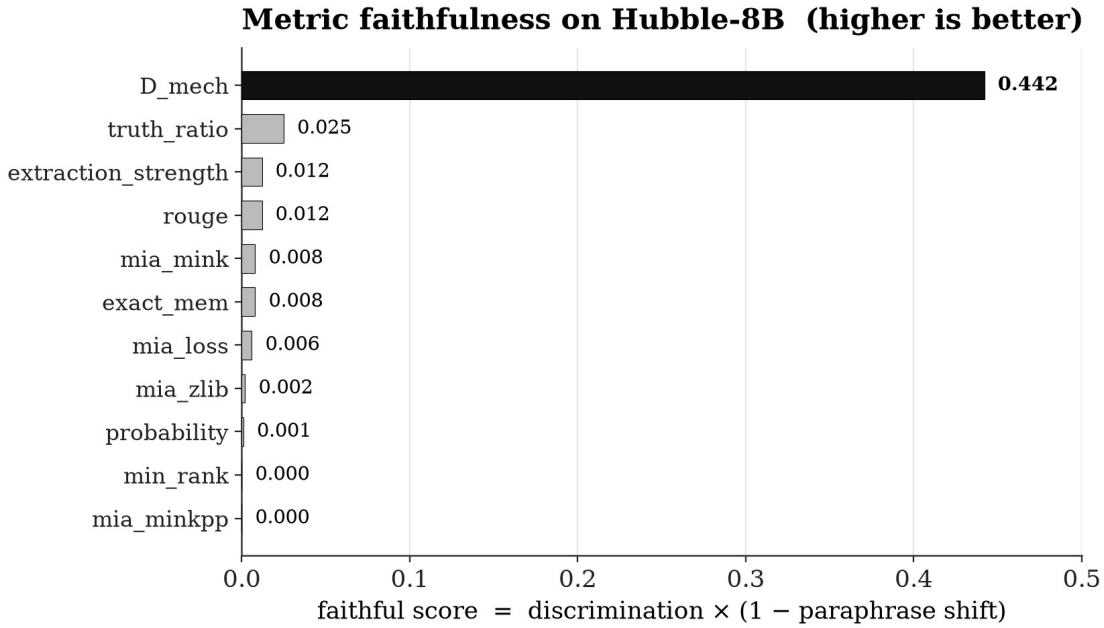
discrim, how cleanly KNOWS separates from ABSENT (1 = perfect, higher is better)

– |Cliff's δ| between the P pool and the N pool. Most metrics score high here, discrimination is easy.

– A metric is faithful only if it has BOTH: low invar_shift AND high discrim. Only D_mech does.

only the mechanistic metric is faithful





- Discrimination is easy, nearly every metric separates knows from absent. The real test is invariance.
- Only D_mech stays invariant under paraphrase (shift 0.47 vs. ≥ 0.95); its faithfulness score is ~18× the best lexical metric.
- It is the DISTANCE construction that wins, naive min-rank, also mechanistic, fails invariance.

logit, tuned, and Jacobian readouts

– A transformer refines its answer across layers in a residual stream. A "lens" transports one middle layer's state into output-token space so we can read the rank of the answer there. The three lenses differ only in that transport.

Logit lens, the model's frozen output head

– Apply the network's own final unembedding to a middle layer, as if it stopped there. Parameter-free and fully reproducible; but early layers aren't yet in "output coordinates," so it reads noisy there.

Tuned lens, a trained per-layer correction (Belrose 2023)

– Learn a small affine map per layer so the readout reproduces the model's FINAL distribution (trained on text; here to KL/layer ≈ 0.05). Cleaner mid-stack, but because it is trained on the output, it "skips ahead."

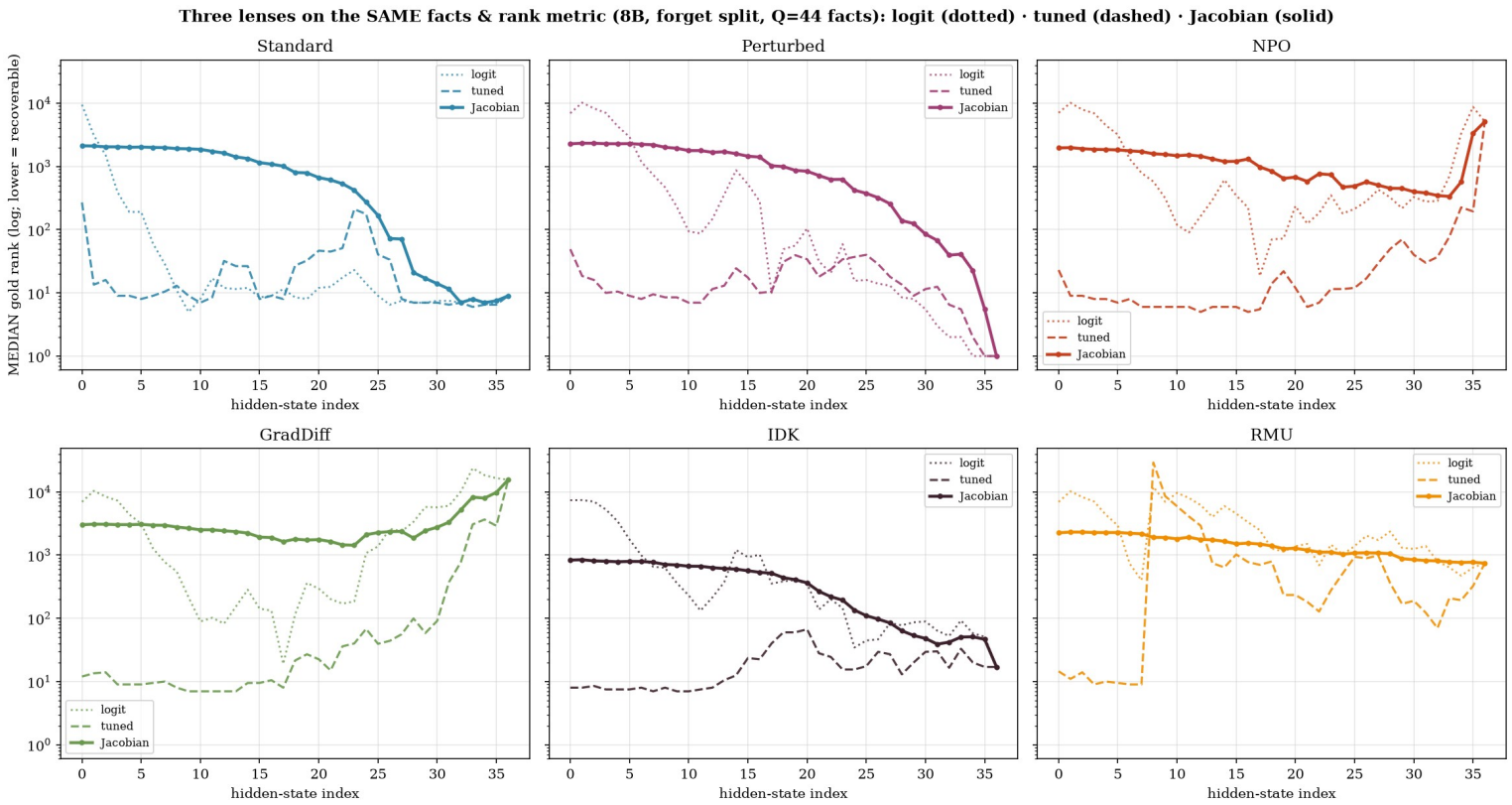
Jacobian lens, the model's own gradient map (Anthropic 2026)

– Read the layer through A_L = average of ∂logits/∂h over diverse text (plus an intercept): a first-order view of the network's actual tail. Causal, it reflects what the model does with the state, not a fitted correlation. Costly, conservative.

The ladder

– Same internal state, three transports, correlational (logit) → calibrated (tuned) → causal (Jacobian).

the same 44 forget facts, one rank metric (8B)



– Tuned (dashed) sits at rank ~10 from LAYER 0 for every model, even suppressed NPO / GradDiff. It "skips ahead," laundering the answer into every layer and hiding both where computation happens and the suppression itself. So it is NOT more faithful for our question.

– Logit (dotted): genuine emergence, buried early, surfaces late. Jacobian (solid): smooth and causal, reproduces the taxonomy (Perturbed → 1, Standard → 9, NPO / GradDiff buried) and shows the late-layer RISE = active suppression. For NPO it keeps the fact buried at every layer even though the logit lens still sees a lingering linear trace (causal ≠ correlational).

the metric needs a reproducible, shared, signal-preserving readout

purpose	best lens	why
Base for the D_mech metric	Logit	parameter-free, reproducible, preserves the trajectory signal
Qualitative causal claim	Jacobian	reflects what the model actually computes
Predict the eventual output early	Tuned	calibrated, but useless for where / whether

– D_mech must be a property of the MODEL, not our choices, the logit lens is the only readout with no trained or estimated parameters (tuned depends on training; Jacobian on the reference set), so the metric is exactly reproducible.

– A tuned-based D_mech would FAIL: tuned flattens every trajectory to rank ~10 (NPO ≈ Perturbed mid-stack), erasing the knows-vs-suppressed signal D_mech reads.

– Invariance needs ONE shared, fixed transport so rewording cancels across the two anchors, only the logit lens is fixed; and D_mech's band (layers 18–36) already avoids the logit lens's one weakness (early-layer noise).

– Verdict: logit-lens D_mech is proven faithful, parameter-free, reproducible. Tuned = cross-check; Jacobian = causal confirmation, neither is a better metric base.

remaining directions

A larger averaged-Jacobian transport

– Our causal lens averages over a modest reference set. The full Anthropic (2026) averaged-transport pipeline (large data, their normalization) should sharpen it, and may recover the knows-vs-never-knew separation at 1B.

From observation to causation

– Causal patching / activation steering of the suppressed representation, to turn "suppression ≠ erasure" from an observation into a causal intervention result.

Robustness of the faithful metric

– Multi-paraphrase and 1B robustness of D_mech, more wrong-target realizations, and larger fact sets.

CONCLUSION

Contribution

We add a mechanistic-interpretability metric to the Hubble unlearning benchmark and show, under the field's own meta-evaluation, that it is the only metric **that is both knowledge-discriminative AND paraphrase-invariant,** empirically confirming that output and lexical metrics confuse surface form with knowledge.

Along the way: a corrected logit-lens readout, and a scale-dependent method verdict (NPO most scale-robust; GradDiff/RMU work at 8B not 1B; IDK surface at both).